P1-06 — Formal Moment Analysis for C5 and C6

Command: P1-06 — Formal Moment Analysis for C5 and C6 **Date:** 21 October 2026 **Inputs (all previously proved):** T1–T6, C1–C4 (P1_04_PROOFS_T1_T4.md, P1_05_PROOFS_T5_T6_COROLLARIES.md). **Frozen statements:** transcribed below before any derivation; 02_THEOREM_BLUEPRINT.md is **not** edited. **PC-1 remains unapproved**: the corrected C5 hull mean is proved here as mathematics and presented as a proposal; adopting it into the blueprint requires the separate approval command.

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0. Frozen statements (transcribed) and the random model

Frozen C5. “Uniform random labeled assignment par E[dim C] = (n/2)·(Σ_s m_s) (kul F_q-lambai ka aadha) aur E[Hull] = [P(2P+1)/(6(P+1))] · Σ_O w_O L_O. Chaabi: ek kenare par min(P−α_i, α_{i−1}) do swatantra uniform {0,…,P} ka min hai; E[min] = Σ_{t=1}^P ((P−t+1)/(P+1))² = P(2P+1)/(6(P+1)). P=1 par 1/4 — purana mean wapas.”

Frozen C6. “Variance: Orbit-swatantra yog; L=1 (self-loop: min(P−A,A)), L=2 (do kenare ek hi jodi ko saanjha karte hain), L≥3 (sirf padosi kenare sah-sambandhit) — teeno cases ke band sutra (closed forms) Phase 4 me P ke parimaani (polynomial) roop me nikale jayenge.” **[RR-8]**

Status of the frozen text, in advance. The code-dimension mean in C5 is correct (proved in §2.1). The frozen hull mean is **incorrect**: its proof key (“two independent uniforms”) fails exactly on L_O = 1 orbits, where the edge is a self-loop min(P−A, A); consequently the “P=1 gives 1/4” remark is also wrong for those orbits (ν_1 = 0, §2.2). A corrected statement is proved in §2.4 and registered as proposal PC-1 (unapproved). C6 as frozen asserts no formula; §3 supplies the closed forms, discharging RR-8 at the level of mathematics (blueprint adoption again awaits the approval command).

Random model. A labeled assignment is chosen uniformly from the finite product set of T1/P1-05 Proposition 3.1. Equivalently: for each component s, orbit O ∈ O_s, and position i, the exponent A^{(s,O)}_i is uniform on {0, …, P}, and all these variables (over all s, O, i) are mutually independent — uniformity on a Cartesian product is the product of the uniform measures. For one orbit, drop the superscripts, write L = L_O, w = w_O, and define the cyclic edge variables

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X_i = min(P − A_i, A_{i−1}),   i = 0, …, L−1   (indices mod L),     S_L = Σ_{i=0}^{L−1} X_i.
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By T4 (P1-04/P1-05), the orbit contributes w·(P − A_i)-type terms to the code dimension and w·S_L to dim_{F_q} Hull_k; globally,

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dim_{F_q} Hull_k = Σ_{s} Σ_{O∈O_s} w_O · S_{L_O}^{(s,O)}.            (1)
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All expectations/variances below are exact rational numbers.

Why L = 1 and L ≥ 2 must differ. For L ≥ 2, the two arguments of each edge minimum, P − A_i and A_{i−1}, come from **two distinct** independent uniforms (also for L = 2: the two edges are the two ordered pairings of the same two distinct variables). For L = 1 the single edge is a **self-loop**: its two arguments are P − A and A, two functions of the ∗same∗ variable. The statistics are genuinely different — e.g. at P = 1, min(1−A, A) ≡ 0 on {0,1} (mean 0), whereas min of two independent bits has mean 1/4.

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1. One-edge moments (for L ≥ 2 edges)

Let U, B be independent uniform on {0, …, P} and X = min(U, B) (an L ≥ 2 edge has this law, with U = P − A_i).

Lemma 1.1 (edge mean; frozen C5’s key quantity). E[X] = μ_P := P(2P+1)/(6(P+1)).

Proof. For integer-valued X ≥ 0, E[X] = Σ_{t≥1} Pr(X ≥ t). Now Pr(min(U,B) ≥ t) = ((P−t+1)/(P+1))² for 1 ≤ t ≤ P, so E[X] = (1/(P+1)²) Σ_{t=1}^P (P−t+1)² = (1/(P+1)²) Σ_{r=1}^P r² = (1/(P+1)²)·P(P+1)(2P+1)/6 = P(2P+1)/(6(P+1)). ∎

Lemma 1.2 (edge second moment and variance).

$$E[X^2] = P(P^2+P+1)/(6(P+1)), \quad \mathrm{Var}(X) = P(P+2)(2P^2+4P+3) / (36(P+1)^2).$$

Proof. For integer-valued $X \geq 0$, $E[X^2] = \Sigma_{t\geq 1} (2t-1) \Pr(X \geq t)$ (since $x^2 = \Sigma_{t=1}^{x} (2t-1)$). With the same tail probabilities and the substitution $r = P-t+1$: $E[X^2] = (1/(P+1)^2) \Sigma_{r=1}^{P} (2P+1-2r)\, r^2 = (1/(P+1)^2) [\, (2P+1)\Sigma r^2 - 2\Sigma r^3 \,]$. Using $\Sigma_{r=1}^{P} r^2 = P(P+1)(2P+1)/6$ and $\Sigma_{r=1}^{P} r^3 = (P(P+1)/2)^2$, the bracket equals $P(P+1)(2P+1)^2/6 - P^2(P+1)^2/2$; dividing by $(P+1)^2$ and factoring the bracket as $P(P+1)\cdot[\, (2P+1)^2 - 3P(P+1) \,]/6 = P(P+1)(P^2+P+1)/6$ (since $(2P+1)^2 - 3P(P+1) = P^2+P+1$) gives $E[X^2] = P(P^2+P+1)/(6(P+1))$. Then $\mathrm{Var}(X) = E[X^2] - \mu_P^2 = [6P(P+1)(P^2+P+1) - P^2(2P+1)^2] / (36(P+1)^2)$, and the numerator expands (equivalently, factors) as $P(P+2)(2P^2+4P+3)$: indeed $6P(P+1)(P^2+P+1) - P^2(2P+1)^2 = 2P^4 + 8P^3 + 11P^2 + 6P = P(P+2)(2P^2+4P+3)$, as is checked by expanding the right side. ∎

Lemma 1.3 (the one-variable profile). For $0 \leq c \leq P$ and U uniform on $\{0,\ldots,P\}$, $F(c) := E[\min(c, U)] = c(2P-c+1)/(2(P+1))$.

Proof. $E[\min(c,U)] = \Sigma_{t=1}^{c} \Pr(U \geq t) = \Sigma_{t=1}^{c} (P-t+1)/(P+1) = (1/(P+1))\cdot[\, cP - c(c-1)/2 \,] = c(2P-c+1)/(2(P+1))$. ∎

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2. C5 — Means

2.1 Code-dimension mean (frozen C5, first part: correct)

For each factor f, $E[P - A_f] = P - P/2 = P/2$ (mean of the uniform on $\{0,\ldots,P\}$ is $P/2$). By T4 and linearity, $E[\dim_{F_q} C] = \Sigma_s m_s \Sigma_{f\in F_s} \deg(f)\cdot(P/2) = \Sigma_s m_s \cdot n_0 \cdot (P/2) = (n/2)\cdot\Sigma_s m_s$, using $\Sigma_{f\in F_s} \deg f = \deg S_s = n_0$ and $n = P\, n_0$. ∎ (This is exactly the frozen claim; no correction needed.)

2.2 The L = 1 orbit mean

Proposition 2.2. For $L = 1$, $E[X_0] = E[\min(P-A, A)] = \nu_P := \lfloor P^2/4 \rfloor/(P+1)$.

Proof. $E[\min(P-A,A)] = (1/(P+1)) \Sigma_{a=0}^{P} \min(P-a, a)$. The summands are symmetric under $a \mapsto P-a$. If $P = 2m$: the values are $0, 1, \ldots, m-1, m, m-1, \ldots, 0$, summing to $2\cdot\Sigma_{a=0}^{m-1} a + m = m(m-1) + m = m^2 = P^2/4$. If $P = 2m+1$: the values are $0, 1, \ldots, m, m, \ldots, 0$, summing to $2\cdot\Sigma_{a=0}^{m} a = m(m+1) = (P^2-1)/4$. Both equal $\lfloor P^2/4 \rfloor$; divide by $P+1$. ∎

In particular $\nu_1 = 0$, $\nu_2 = 1/3$, $\nu_3 = 1/2$ — and $\nu_P \neq \mu_P$ for every $P \geq 1$ (e.g. $\mu_P - \nu_P > 0$ since $\mu_P \geq \mu_1 = 1/4 > 0 = \nu_1$ at $P=1$, and directly from the formulas in general), which is the quantitative form of the self-loop phenomenon of §0.

2.3 The L ≥ 2 orbit mean

Proposition 2.3. For $L \geq 2$, $E[S_L] = L\cdot\mu_P$.

Proof. Each X_i has the law of Lemma 1.1: for $L \geq 3$, A_i and A_{i-1} are distinct coordinates; for $L = 2$, $X_0 = \min(P-A_0, A_1)$ and $X_1 = \min(P-A_1, A_0)$ are the two ordered pairings of the distinct pair (A_0, A_1), each marginally min of two independent uniforms. Hence $E[X_i] = \mu_P$ for all i, and linearity of expectation (no independence needed) gives $E[S_L] = L\,\mu_P$. ∎

2.4 Global mean and the PC-1 proposal

Theorem 2.4 (mean hull dimension; proved mathematics). Under (G1)–(G6) of P1-05 and the uniform labeled model,

$$E[\dim_{F_q} \mathrm{Hull}_k] = \nu_P \cdot \Sigma_{s} \Sigma_{O\in O_s,\, L_O = 1} w_O \;+\; \mu_P \cdot \Sigma_{s} \Sigma_{O\in O_s,\, L_O \geq 2} w_O\cdot L_O.$$

Proof. Take expectations in (1): $E[\dim \mathrm{Hull}] = \Sigma_{s,O} w_O\cdot E[S_{L_O}]$ by linearity, and apply Proposition 2.2 ($L_O = 1$) and Proposition 2.3 ($L_O \geq 2$) orbit by orbit. (Cross-check via the proved enumerator: $E[\dim \mathrm{Hull}] = [z\cdot\partial_z E(u,z)/E(u,z)]_{u=z=1}$ by T6, and the logarithmic derivative splits over the orbit product into the same per-orbit terms; the two computations agree because both are linearity of expectation over the orbit decomposition.) ∎

Proposal PC-1 (unapproved — precise change requested). Replace the frozen C5 hull-mean display E[Hull] = [P(2P+1)/(6(P+1))]·Σ_O w_O L_O and its proof key by Theorem 2.4's split formula. What changes and why:

Frozen C5 piece	Status	Replacement / reason
E[dim C] = (n/2)·Σ_s m_s	**Correct** (§2.1)	Unchanged.
E[Hull] = μ_P·Σ_O w_O L_O	**Incorrect** whenever an L_O = 1 orbit exists	Split sum: ν_P on L_O = 1 orbits, μ_P·L_O on L_O ≥ 2 orbits (Theorem 2.4). The frozen proof key treats the self-loop edge as a min of two ∗independent∗ uniforms; it is min(P−A, A).
"P=1 par 1/4 — purana mean wapas"	**Incorrect** for L_O = 1	At P = 1: E[Hull] = (1/4)·Σ_{L_O ≥ 2} w_O L_O (since ν_1 = 0, μ_1 = 1/4); L_O = 1 orbits contribute 0.

Counterexamples to the frozen display (already brute-forced at vector level in P1-03 and reproduced by validate_p1_06.py at the orbit level): any model with an L = 1 orbit, e.g. P = 2, L = 1: true per-orbit mean ν_2 = 1/3 vs frozen μ_2 = 5/9. **This proposal changes nothing until approved; the frozen file stands.**

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3. C6 — Variance of the hull dimension (RR-8)

Global structure first. In (1), the summands w_O·S_{L_O}^{(s,O)} for distinct (s, O) are functions of **disjoint** subfamilies of the independent exponent variables, hence are mutually independent under the product-uniform model (this is where CRT/product independence enters: the assignment set is a Cartesian product, T1/P1-05 Prop. 3.1, so the uniform measure factorizes over components and orbits). Therefore

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Var(dim_{F_q} Hull_k) = Σ_{s} Σ_{O∈O_s} w_O² · Var(S_{L_O}),          (2)
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and it suffices to compute Var(S_L) for one orbit. Note the variance adds with squared weights w_O², the mean (Theorem 2.4) with weights w_O.

3.1 L = 1: the self-loop variance

Proposition 3.1. With X = min(P−A, A), A uniform on {0,...,P}:

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E[X²] = P(P²+2)/(12(P+1))  if P even;    E[X²] = P(P−1)/12  if P odd;
Var(X) = P(P+2)(P²+2P+4)/(48(P+1)²)  if P even;    Var(X) = (P−1)(P+3)/48  if P odd.
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Proof. E[X²] = (1/(P+1)) Σ_{a=0}^P min(P−a, a)². For P = 2m the squared values are 0²,..., (m−1)², m², (m−1)²,..., 0², so the sum is 2·Σ_{a=0}^{m−1} a² + m² = m(m−1)(2m−1)/3 + m² = m(2m²+1)/3; substituting m = P/2 gives E[X²] = P(P²+2)/(12(P+1)). For P = 2m+1 the values 0,...,m each occur twice, so the sum is 2·Σ_{a=0}^{m} a² = m(m+1)(2m+1)/3 = P(P²−1)/12 (using m = (P−1)/2, m+1 = (P+1)/2, 2m+1 = P); dividing by P+1 gives E[X²] = P(P−1)/12. Subtracting ν_P² (Proposition 2.2: ν_P = P²/(4(P+1)) for even P in the form m²/(2m+1), and ν_P = (P−1)/4 for odd P): for even P, Var = P(P²+2)/(12(P+1)) − P⁴/(16(P+1)²) = P(P+2)(P²+2P+4)/(48(P+1)²) (common denominator 48(P+1)²; the numerator identity 4P(P²+2)(P+1) − 3P⁴ = P(P+2)(P²+2P+4) is by expansion); for odd P, Var = P(P−1)/12 − ((P−1)/4)² = (P−1)(P+3)/48 (factoring (P−1) and using P/12 − (P−1)/16 = (P+3)/48). ∎

(Checks: P = 1: Var = 0, indeed X ≡ 0. P = 2: 2/9. P = 3: 1/4.)

3.2 L ≥ 3: adjacent covariance, independent non-neighbours

For L ≥ 3, X_i is a function of the pair (A_i, A_{i−1}).

Lemma 3.2 (adjacent covariance). For each i, with X_i, X_{i+1} consecutive edge variables (indices mod L, L ≥ 3),

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Cov(X_i, X_{i+1}) =: κ_P = − P(P+2)(7P²+14P+9) / (360(P+1)²)   (< 0 for P ≥ 1).
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Proof. X_i = min(P−A_i, A_{i−1}) and X_{i+1} = min(P−A_{i+1}, A_i) share exactly the one variable A_i (for L ≥ 3; in particular this includes every pair when L = 3). Condition on A_i = b: then X_i = min(P−b, A_{i−1}) and X_{i+1} = min(P−A_{i+1}, b) are functions of the distinct variables A_{i−1}, A_{i+1}, hence conditionally independent, with conditional means E[min(P−b, A_{i−1})] = F(P−b)

and E[min(P−A_{i+1}, b)] = F(b) (Lemma 1.3, using P−A_{i+1} uniform). Therefore E[X_i X_{i+1}] = E[F(P−A_i) F(A_i)] = (1/(P+1)) Σ_{b=0}^P F(P−b) F(b). Substituting F(c) = c(2P−c+1)/(2(P+1)): the product in the sum is b(P−b)(2P+1−b)(P+1+b) / (4(P+1)²), and b(P−b)(2P+1−b)(P+1+b) = P(2P+1)(P+1)·b − (P²+3P+1)·b² − 2P·b³ + b⁴ (expansion). Summing with Σ_{b=0}^P b = P(P+1)/2, Σ b² = P(P+1)(2P+1)/6, Σ b³ = (P(P+1)/2)², Σ b⁴ = P(P+1)(2P+1)(3P²+3P−1)/30 and simplifying gives E[X_i X_{i+1}] = P(P−1)(11P²+15P+6)/(120(P+1)²). (At P = 1 this is 0, correctly: X_i = 1 forces A_i = 0, which forces X_{i+1} = 0.) Finally κ_P = E[X_iX_{i+1}] − μ_P²: with common denominator 360(P+1)², the numerator of E[X_iX_{i+1}] is 3P(P−1)(11P²+15P+6) and the numerator of μ_P² is 10P²(2P+1)²; their difference is −7P⁴ − 28P³ − 37P² − 18P = −P(P+2)(7P²+14P+9) (both expansions agree), so κ_P = −P(P+2)(7P²+14P+9)/(360(P+1)²) as displayed. The sign is negative for P ≥ 1 since every factor in P(P+2)(7P²+14P+9) is positive — intuitively, a large shared A_i shrinks P−A_i inside X_i while enlarging the cap A_i inside X_{i+1}. ∎

Lemma 3.3 (which pairs are independent). For L ≥ 4, if j ∉ {i−1, i, i+1} (mod L), then X_i and X_j are independent, hence Cov(X_i, X_j) = 0. The L unordered adjacent pairs {i, i+1} are exactly the pairs with possibly nonzero covariance (for L = 3 these are all 3 unordered pairs).

Proof. X_i is a function of {A_i, A_{i−1}} and X_j of {A_j, A_{j−1}}; the stated condition is exactly that these two 2-element index sets are disjoint. Functions of disjoint subfamilies of independent random variables are independent. Conversely each adjacent pair shares exactly one variable (Lemma 3.2's setup). ∎

Proposition 3.4 (L ≥ 3 variance). Var(S_L) = L · P(P+2)(P²+2P+2) / (60(P+1)²).

Proof. Var(S_L) = Σ_i Var(X_i) + 2·Σ_{i<j} Cov(X_i, X_j). Every X_i has variance Var(X) of Lemma 1.2 (marginally an L ≥ 2 edge). By Lemmas 3.2–3.3, among the unordered pairs exactly the L adjacent pairs contribute, each with covariance κ_P. Hence Var(S_L) = L·Var(X) + 2L·κ_P = L·[P(P+2)(2P²+4P+3)/(36(P+1)²) − P(P+2)(7P²+14P+9)/(180(P+1)²)]. With denominator 180(P+1)² the bracket's numerator is P(P+2)·[5(2P²+4P+3) − (7P²+14P+9)] = P(P+2)(3P²+6P+6) = 3P(P+2)(P²+2P+2), so the bracket equals P(P+2)(P²+2P+2)/(60(P+1)²). ∎

3.3 L = 2: the two edges share both variables

For L = 2, X_0 = min(P−A_0, A_1) and X_1 = min(P−A_1, A_0): the pair is not an "adjacent with one shared variable" pair — both variables appear in both edges, and Lemma 3.2 does not apply.

Lemma 3.5 (L = 2 cross moment). E[X_0 X_1] = P²(P−1)/(12(P+1)), and Cov(X_0, X_1) = −P²(P+2)²/(36(P+1)²).

Proof. Write A, B for A_0, A_1; X_0 X_1 = min(P−A, B)·min(P−B, A). For nonnegative integers, E[X_0X_1] = Σ_{s,t ≥ 1} Pr(X_0 ≥ s, X_1 ≥ t). Now X_0 ≥ s iff A ≤ P−s and B ≥ s; X_1 ≥ t iff B ≤ P−t and A ≥ t. Both hold iff t ≤ A ≤ P−s and s ≤ B ≤ P−t, which is nonempty iff s + t ≤ P, in which case each of A, B has exactly P−s−t+1 admissible values. Hence Pr = ((P−s−t+1)/(P+1))² if s+t ≤ P, else 0. Grouping by r = P−s−t+1 ∈ {1, ..., P−1} (the number of positive pairs (s,t) with s+t = P−r+1 is P−r): E[X_0X_1] = (1/(P+1)²) Σ_{r=1}^{P−1} (P−r) r² = (1/(P+1)²)·P²(P−1)(P+1)/12 = P²(P−1)/(12(P+1)), where Σ_{r=1}^{P−1} (P−r)r² = P·(P−1)P(2P−1)/6 − ((P−1)P/2)² = P²(P−1)(P+1)/12 (using the power sums up to P−1). Then Cov = E[X_0X_1] − μ_P²; with denominator 36(P+1)², E[X_0X_1] = 3P²(P²−1)/(36(P+1)²), and 3(P²−1) − (2P+1)² = 3P²−3 − (4P²+4P+1) = −P²−4P−4 = −(P+2)², giving Cov(X_0,X_1) = −P²(P+2)²/(36(P+1)²). ∎

Proposition 3.6 (L = 2 variance). Var(S_2) = P(P+2)(P²+2P+3) / (18(P+1)²).

Proof. Var(S_2) = 2·Var(X) + 2·Cov(X_0,X_1) (Lemma 1.2, Lemma 3.5) = (2/(36(P+1)²))·[P(P+2)(2P²+4P+3) − P²(P+2)²]. The bracket factors: P(P+2)·[(2P²+4P+3) − P(P+2)] = P(P+2)(P²+2P+3). Hence Var(S_2) = P(P+2)(P²+2P+3)/(18(P+1)²). ∎

Sanity: at P = 1, S_2 ∈ {0, 1} with Pr(S_2 = 1) = 1/2, so Var = 1/4; the formula gives 1·3·6/(18·4) = 1/4. Note the L = 2 formula is **not** the L = 2 substitution of Proposition 3.4 (which would give 2·P(P+2)(P²+2P+2)/(60(P+1)²); at P = 2: 8/27 ≈ 0.296 vs the correct 44/81 ≈ 0.543) — the shared-pair dependence is genuinely different, as frozen C6 anticipated.

3.4 Global variance theorem (C6 mathematics)

Theorem 3.7 (variance of the hull dimension). Under the hypotheses of Theorem 2.4,

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Var(dim_{F_q} Hull_k) = Σ_{s} Σ_{O∈O_s} w_O² · V_{L_O}(P),
V_1(P) = P(P+2)(P²+2P+4)/(48(P+1)²)  (P even),   (P−1)(P+3)/48  (P odd);
V_2(P) = P(P+2)(P²+2P+3)/(18(P+1)²);
V_L(P) = L·P(P+2)(P²+2P+2)/(60(P+1)²)   (L ≥ 3).
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Proof. Equation (2) (independence across components/orbits under the product-uniform model, via T1 and P1-05 Proposition 3.1) reduces the claim to Var(S_L) for one orbit, which is Proposition 3.1 (L = 1), Proposition 3.6 (L = 2), and Proposition 3.4 (L ≥ 3). The squared weight w_O² appears because Var(w_O·S) = w_O² Var(S). No cross-orbit covariance terms appear: distinct orbits involve disjoint exponent variables. ∎

What remains unsimplified / open. Nothing in the variance of dim Hull is left as a finite sum: every case is a closed rational function of P (piecewise by parity only for L = 1). Not treated here (and not claimed): variance of dim C, mixed covariance Cov(dim C, dim Hull), higher moments, and the variance under any non-uniform or equivalence-class model. Blueprint adoption of Theorem 3.7 as C6's closed forms is part of the same future approval decision as PC-1…PC-6 (PC-6 asked that C6 be marked "to be proved (RR-8)" until then; RR-8 is now discharged mathematically).

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4. Dependency map

Previously proved: T1 (assignment product model), T4 (hull dimension = orbit sums (1)), T6 (enumerator; used only for the cross-check remark in Theorem 2.4), P1-05 Prop. 3.1 (component/orbit independence of the product model). New in this package: Lemmas 1.1–1.3 (edge moments), Propositions 2.2–2.3 and Theorem 2.4 (means; PC-1 mathematics), Propositions 3.1/3.4/3.6, Lemmas 3.2/3.3/3.5, Theorem 3.7 (variances; RR-8 mathematics). Still open elsewhere: C7 (RR-9), PC-1…PC-6 approvals, Gate G1 (novelty; untouched).

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5. Computational verification (supporting evidence, not proof)

Executed for this package (21 Oct 2026); earlier scripts/outputs unmodified:

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python3 code/validate_p1_06.py    # NEW; output code/validation_p1_06_output.txt
                                  # final line: ALL P1-06 MOMENT CHECKS PASSED
python3 code/validate_p1_03.py    # rerun: exit 0, byte-identical to saved output
python3 code/validate_p1_05.py    # rerun: ALL P1-05 MULTI-COMPONENT ASSEMBLY CHECKS PASSED
```

validate_p1_06.py works in exact rational arithmetic and compares three layers for P ∈ {1, 2, 3, 4, 5, 8, 9} (all prime powers p^ϱ, including P = 1 = p^0): (1) the closed forms of §§1–3; (2) exact finite sums for every building block (Σ_a min(P−a,a)^{1,2}, edge tail sums, the Σ_b F(P−b)F(b) adjacent sum, the (s,t) double sum for L = 2); (3) direct enumeration of A-tuples for L ∈ {1, 2, 3, 4, 6} (capped at 300k tuples; the two largest (P, L) = (8,6), (9,6) cases are covered by layers (1)–(2) only). It also verifies the **global** assembly: mean adds with weights w_O, variance with w_O² — against direct global enumeration for P = 2, orbits (L,w) = (1,2), (2,1), (3,1) (729 tuples) and against exact convolution of per-orbit distributions for P = 3, orbits (1,2), (2,1), (3,3), (6,1).

What this verifies: the displayed rational formulas agree with brute force and with the independent finite sums at the tested (P, L), including all three variance regimes and the parity split at L = 1. What it cannot prove: the formulas for untested (P, L) (the proofs in §§1–3 do that), the product-model independence (proved from T1/CRT, §0), or anything outside the labeled uniform compatible-twist model.